

Ipc is the Hosoya index: a numerically stable, size-free reformulation of molecular information content

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Abstract

The information content index **Ipc**, available in standard cheminformatics toolkits, is shown to be both definitionally redundant and numerically unsound. Over 400 molecules, 99.79% of the variance of $\log_2 \text{Ipc}$ is explained by $\log_2 S$ alone, where S is the Hosoya index; the entropy factor contributes 2.24 bits against 32.3. Separately, the Le Verrier–Faddeev–Frame recursion used to obtain the characteristic polynomial loses all significant digits above approximately 80 atoms: for a 120-atom chain the exact final coefficient is +1 and the computed value is -1.97×10^{26} , without overflow or warning. We give an eigenvalue-based algorithm with magnitude-ordered root expansion, accurate to 2.3×10^{-15} at $n = 200$ and eight times faster. We then define $\text{dIpc} = \log_2 Z(G) - \log_2 F(n+1)$, matching information relative to the n -alkane of equal size, whose correlation with atom count is +0.049 against +0.993 for $\log_2 Z$. The upper reference $Z_{\max}$ over chemical graphs is determined exhaustively to $n = 16$, by local search to $n = 24$, and by a fitted law accurate to 0.022 bits thereafter. Applied to 6,121 COCONUT terpenoids against size-matched controls, terpenoid topological variance is half that of other natural products (ratio 2.05, $p = 1.4 \times 10^{-146}$), consistently across all size strata.

Introduction

Molecular descriptor blocks are computed on every molecule in every model that uses them, so a defective descriptor propagates silently and at scale. RDKit’s `descList` contains 217 entries, two of which — **Ipc** and **AvgIpc** — are information-theoretic indices over the characteristic polynomial of the adjacency matrix [1,2].

This work asks whether those two indices measure what they are intended to measure, and whether they are computed correctly. Both answers are negative. We identify the quantity **Ipc** actually tracks, give a corrected algorithm for the underlying polynomial, propose a reformulation with analytic references at both ends of its range, and apply it to natural product chemical space.

Methods

Exact characteristic polynomial. The adjacency matrix of a molecular graph is integer, so its characteristic polynomial has integer coefficients. We compute them exactly by running the Faddeev–LeVerrier recursion modulo several small primes and reconstructing by the Chinese remainder theorem, with primes chosen so that every modular matrix product remains exactly representable in double precision (S2.1). Results were verified against exact rational arithmetic.

Information quantities. With coefficient magnitudes c , let $S = \sum |c_i|$ and $H = -\sum (c_i/S) \log_2 (c_i/S)$. The standard descriptors are $\text{Ipc} = S \cdot H$ and $\text{AvgIpc} = H$. We evaluate both by log-sum-exp so that S is never formed explicitly.

The dIpc index. For an acyclic graph, S equals the Hosoya index Z , the number of matchings [1]. Among trees on n vertices the path maximises Z , with $Z = F(n+1)$, the Fibonacci number. We therefore define

$$\text{dIpc}(G) = \log_2 Z(G) - \log_2 F(n+1),$$

which is zero for an unbranched chain, negative for branched acyclic graphs, and positive when rings contribute additional matchings.

Upper reference. Since adding an edge cannot reduce the matching count, the maximum over chemical graphs (connected, maximum degree ≤ 4) is attained on 4-regular graphs. These were enumerated exhaustively with **geng** from *nauty* [5] and their matchings counted by subset dynamic programming. Beyond $n = 16$ we used local search over 2-opt edge swaps with an exact objective, which yields lower bounds (S2.2); beyond $n = 24$ the dynamic program exceeds memory and we use a fitted law bracketed by an explicit construction below and a density ceiling above.

Natural product data. COCONUT 2022.01.01 structures with NPClassifier pathway labels [6,7,8]. Terpenoids are those with pathway exactly **Terpenoids** and one of five terpene superclasses; compounds labelled **Alkaloids**, **Terpenoids** are excluded. Controls are single-pathway non-terpenoids matched one-for-one on heavy-atom count, since the spread of dIpc grows with molecular size and a range restriction would leave a confounding gradient. The largest fragment is used throughout, as Z is multiplicative over components.

Results

Ipc is dominated by the Hosoya index

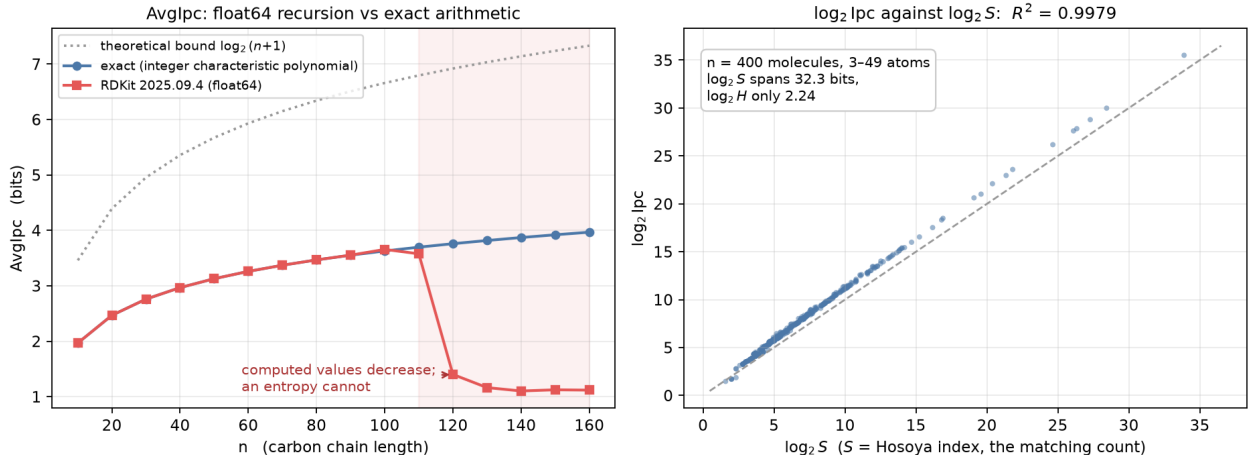


Figure 1: **Left:** AvgIpc from RDKit 2025.09.4 compared with exact integer arithmetic for linear carbon chains, $n = 10$ –160; dotted line is the bound $\log_2(n+1)$. **Right:** $\log_2$ Ipc against $\log_2 S$ for 400 molecules of 3–49 heavy atoms; dashed line is $y = x$.

The Bonchev–Trinajstić total information content $I = N \cdot H$ is defined for a population of N elements partitioned into classes by H [2]. In $I_{pc} = S \cdot H$, however, H is an entropy over the $n+1$ polynomial coefficients, bounded by $\log_2(n+1)$, while S counts matchings and grows as φ^n . The two factors refer to different populations.

Empirically the product is dominated by S . Across 400 molecules, $\log_2 S$ spans 32.3 bits and $\log_2 H$ spans 2.24, with $R^2 = 0.9979$ between $\log_2 I_{pc}$ and $\log_2 S$ (Figure 1, right). As $\log_2 S$ correlates with heavy-atom count at $r = 0.993$, Ipc is approximately a measure of molecular size.

The characteristic polynomial is computed incorrectly above 80 atoms

The Le Verrier–Faddeev–Frame recursion is sequential: coefficient k is formed after k matrix products have grown the intermediates to the magnitude of the largest coefficient, then obtained as a difference of quantities of that magnitude, with each coefficient entering the next matrix.

n	AvgIpc, RDKit	exact	coefficients in error by $> 1\%$
60	3.2594	3.2594	0.0%
80	3.46687	3.46686	—
100	3.6561	3.6278	19.6%
120	1.4007	3.7593	23.0%
250	1.1084	4.2886	—

AvgIpc is a Shannon entropy bounded by $\log_2(n+1)$ and cannot decrease as a chain is extended; it begins decreasing near 110 atoms (Figure 1, left). Boundedness does not confer safety: the quantity is computed from the same corrupted coefficients and degrades earlier and less visibly than the unbounded Ipc. No overflow occurs — at $n = 250$, $S = 1.3 \times 10^{52}$ against a double-precision ceiling of 1.8×10^{308} — so the returned values are finite.

Expanding the polynomial from its eigenvalues, with root factors multiplied in order of increasing magnitude, keeps each partial product near the scale of the final coefficients. Maximum relative error falls from 5.4×10^{15} to 2.3×10^{-15} at $n = 200$, with an eight-fold speed-up. The ordering is essential: unordered expansion still reaches 4.8×10^4 relative error at $n = 160$. The correction has been submitted upstream (rdkit/rdkit#9657).

dIpc removes the size dependence that other normalisations do not

quantity	correlation with heavy-atom count
$\log_2 Z$	+0.993
$\log_2(Z)/n$	+0.409
dIpc	+0.049

Subtracting $\log_2 F(n+1)$ removes the size term exactly, where division by a power of n removes it only approximately. The resulting index is signed and interpretable: zero for an n -alkane, negative with branching, positive with ring fusion. Across 478 acyclic molecules no value exceeded zero, consistent with the extremal property of the path.

$Z_{\max}$ is exact to $n = 16$ and bounded to 0.022 bits beyond

n	$Z_{\max}$	$\log_2 Z/n$	basis
12	2,921	0.95935	exhaustive
14	11,032	0.95924	exhaustive
16	42,439	0.96082	exhaustive
20	598,516	0.95957	local search
24	8,659,688	0.96025	local search
30	4.69×10^8	—	fitted
40	3.66×10^{11}	—	fitted
50	2.86×10^{14}	—	fitted

The fitted law $\log_2 Z_{\max}(n) = 0.96092n - 0.02313$ holds to ± 0.022 bits over $n = 12$ –24. It is bounded below by an explicit construction — a chain of $K_{4,4}$ blocks, each with one internal edge removed so that connector vertices remain within degree 4 — whose exact Z is obtained by transfer matrix at any size and which lies a constant 0.22 bits below the fit up to $n = 248$. It is bounded above by the $K_{4,4}$ density ceiling $0.96342n$. At $n = 50$ the two bounds differ by 0.25 bits.

The density is maximal at $K_{4,4}$ and the sequence shows period-8 structure: a connected 4-regular graph cannot attain the density of disjoint $K_{4,4}$ blocks, as joining two blocks costs 0.0416 bits.

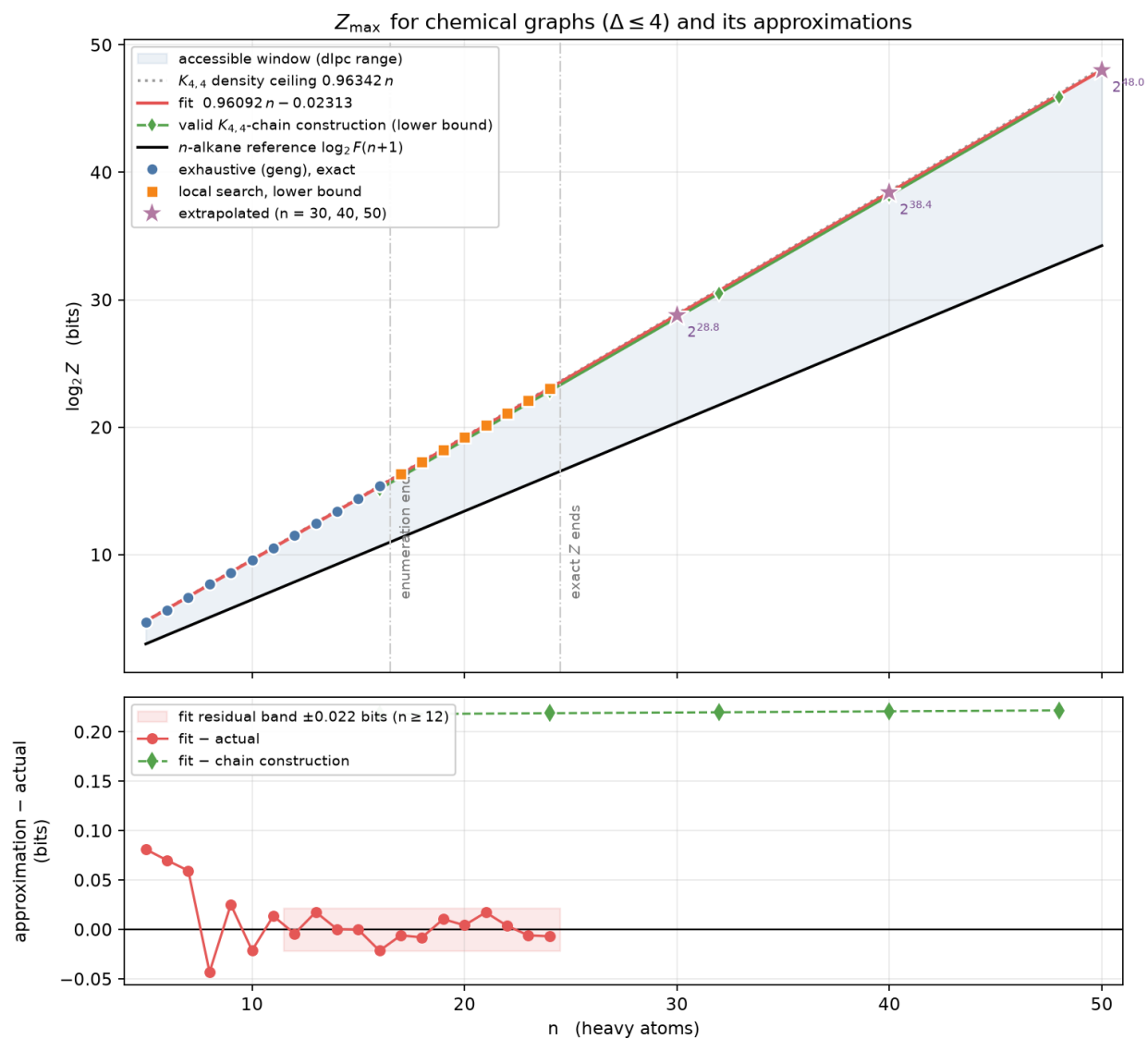


Figure 2: $\log_2 Z$ against n . Blue: exhaustive enumeration, exact. Orange: local search, lower bounds. Green: an explicit $K_{4,4}$ -chain construction. Shaded: the interval between the n -alkane reference and $Z_{\max}$. Lower panel: residuals of the two approximations.

Terpenoid topological variance is half that of other natural products

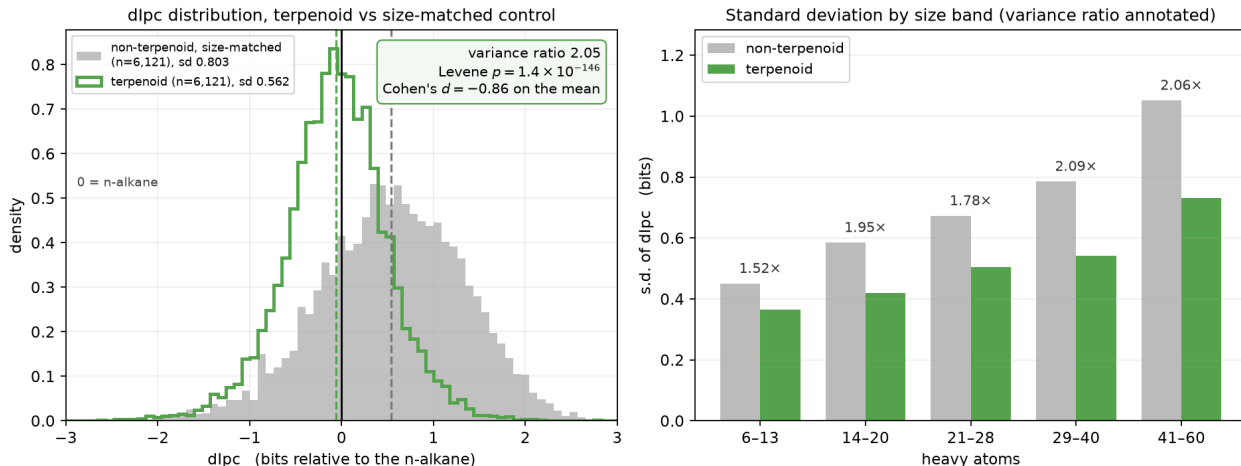


Figure 3: **Left:** dIpc distributions for 6,121 COCONUT terpenoids and 6,121 non-terpenoids matched on heavy-atom count; dashed lines are group means. **Right:** standard deviation by size band, variance ratio annotated.

	terpenoid	non-terpenoid	test
n	6,121	6,121	—
mean heavy atoms	30.72	30.72	matched to 10^{-4}
mean dIpc	-0.058	+0.541	Cohen's $d = -0.86$
s.d.	0.562	0.803	Levene $p = 1.4 \times 10^{-146}$
variance ratio	—	2.045	Fligner $p = 2.8 \times 10^{-146}$

Terpenoid skeletons occupy approximately half the topological variance of size-matched natural products and lie on the branched side of the chain reference, while other pathways lie on the ring-fused side. The restriction is not a size artefact: variance ratios are 1.52, 1.95, 1.78, 2.09 and 2.06 across bands of 6–13, 14–20, 21–28, 29–40 and 41–60 heavy atoms, all with $p < 0.01$.

Both effects are large. The separation in means corresponds to Cohen's $d = -0.86$, and the variance ratio of 2.045 is estimated on 6,121 matched pairs. Terpenoid identity is therefore predictive of position on this axis independently of molecular size.

A weaker secondary trend exists within terpenoids: dIpc decreases by 0.046 ± 0.005 bits per isoprene unit ($r = -0.110$, $p = 4.9 \times 10^{-18}$), explaining about 1% of the variance, and class means are not monotone (S3.3).

Discussion

Ipc should be removed from descriptor blocks. It is approximately a function of molecular size, which is already represented by numerous other descriptors, and it is computed from a polynomial that is unreliable above 80 atoms. Models trained on molecules below about 60 atoms are unaffected by the correction, since old and corrected values agree to 10^{-15} in that range; above it, feature distributions change and should be re-examined rather than simply recomputed.

AvgIpc should be retained once the underlying polynomial is corrected, and dIpc added alongside it. The cost of dIpc is approximately 10 ms per molecule at $n = 50$ using the exact path. It contributes no additional information to a model already provided with $\log_2 Z$ and n and sufficient flexibility to combine them (partial $r = +0.0095$); its value lies in interpretability, in extrapolation beyond the training size range, and in settings

where the model cannot learn the interaction. For a feature matrix, $\log_2 Z$ and n are better supplied as separate columns than pre-combined: division by $n^{1/3}$, $n^{2/3}$ or n adds nothing once both are present (partial r of -0.087 , -0.092 , -0.096).

Terpenoid biosynthesis leaves a measurable signature in graph topology. Assembly from a single branched C5 unit confines terpenoid skeletons to half the topological variance of size-matched natural products and displaces them to the branched side of the chain reference, while pathways not built this way are displaced to the ring-fused side. The effect is large ($d = -0.86$), uniform across the full size range examined, and robust to exact size matching, which is the principal confounder for any index of this kind. dIpc is consequently usable as a size-independent descriptor of biosynthetic class, not only as a repair of Ipc.

Limitations

The isoprene trend within terpenoids explains approximately 1% of the variance and class means are not monotone; it is reported for completeness and no weight is placed on it. Terpenoid assignment derives from a neural classifier rather than curated biosynthetic provenance, and meroterpenoids and steroids were excluded as ambiguous without sensitivity testing. COCONUT 2022.01.01 is a literature aggregation; no deduplication was applied beyond fragment selection, so related scaffolds may be over-represented in both groups equally.

$Z_{\max}$ values for $n = 17$ – 24 are lower bounds rather than proven maxima, with a validated worst-case shortfall of 0.48%; the fitted law is correspondingly a slight underestimate. The optimality of $K_{4,4}$ is confirmed exhaustively at $n = 8$ and $n = 16$ but its asymptotic form is taken from the literature.

dIpc contributes no additional information to a model already supplied with $\log_2 Z$ and n and able to combine them nonlinearly; its value is interpretability and extrapolation, not predictive gain. Its components are established prior results [1,4]; a literature search over OpenAlex returned no previous definition of $\log Z - \log F(n+1)$ as a descriptor, but this is not an exhaustive novelty search.

Retraction. A preliminary analysis of 43 curated terpenes reported no difference in mean dIpc between terpenes and size-matched molecules ($p = 0.106$). That conclusion was incorrect and is retracted: it reflected insufficient power, not absence of effect. The present analysis on 6,121 matched pairs finds a large difference ($d = -0.86$). The same analysis also overestimated the isoprene slope by a factor of 2.5. Its variance-ratio estimate, 2.120, was accurate (S3.3).

References

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Supplementary information

S1. Data

source	count	range	use
boiling-point set	1,000 molecules	2–63 heavy atoms	general chemical space
— variance	400	3–49	Figure 1
decomposition subset			
COCONUT 2022.01.01, NPClassifier-labelled	407,029	—	terpenoid pool
— terpenoids sampled, 5 superclasses	6,121	6–60	Figure 3
— size-matched controls	6,121	6–60	Figure 3
curated terpenes (preliminary)	43 of 46	10–30 carbons	S3.3
connected 4-regular graphs	8,037,418 at $n=16$	$n = 5\text{--}16$	$Z_{\max}$

Terpenoid superclass counts in COCONUT: diterpenoids 23,082; triterpenoids 22,280; sesquiterpenoids 20,418; steroids 16,768; monoterpenoids 6,207; meroterpenoids 3,607; sesterterpenoids 937; carotenoids 953.

Each curated entry was checked against the carbon count its class requires. Three of 46 failed this check and were corrected before use.

S2. Algorithms

S2.1 Modular characteristic polynomial

Primes are selected such that $n(p-1)^2 < 2^{53}$, which keeps each modular matrix product exactly representable in double precision and allows the inner loop to use a floating-point matrix multiply rather than arbitrary-precision integer arithmetic. The number of primes is set from a Hadamard bound on the coefficient magnitudes. Output was verified identical to Faddeev–LeVerrier over exact rationals on random sparse symmetric graphs up to $n = 30$.

S2.2 Local search over 4-regular graphs

Initialisation is a circulant graph $C_n(1, 2)$, which is 4-regular and connected. Each step proposes a 2-opt edge swap: edges (a, b) and (c, d) are deleted and (a, c) , (b, d) inserted, leaving all vertex degrees unchanged. Swaps that disconnect the graph are rejected; the exact matching count is recomputed and the swap accepted when it does not decrease. Multiple restarts from independent random states are used. Every reported value is the exact Z of a realised graph and is therefore a lower bound on $Z_{\max}$.

Validation against exhaustively known maxima: recovered at $n = 10, 12, 13$ and 15 ; at $n = 14$ the best of 24 restarts fell 0.48% short.

S2.3 Block-chain transfer matrix

For blocks joined in a path by single edges, with per-block counts N , N_l , N_r , N_{lr} (total matchings, and those leaving the left, right, or both connectors unmatched),

$$u_{i+1} = u_i N_{lr} + (u_i + m_i) N_r, \quad m_{i+1} = u_i (N_l - N_{lr}) + (u_i + m_i) (N - N_r),$$

where u and m count configurations with the right connector unmatched and matched. This is exact and linear in the number of blocks. An initial version used intact $K_{4,4}$ blocks, which raises connector degree to 5 and violates the valence constraint; one internal edge is therefore removed per block.

S3. Supplementary results

S3.1 Timings

step	n	wall time
exhaustive enumeration and exact Z	10	0.03 s
	12	0.07 s
	14	4.3 s
	15	52 s
	16	1,140 s
local search, 2,000 moves	20	2.1 s
	24	42 s
block-chain transfer matrix	≤ 248	0.17 s

Exhaustive enumeration scales at roughly 10–20 $\times$ per added atom; $n = 17$ would require approximately six hours and $n = 18$ approximately three days.

S3.2 Exhaustive $Z_{\max}$

n	5	6	7	8	9	10	11	12	13	14	15	16
$Z_{\max}$	26	51	100	209	388	780	1482	2921	5600	11032	21482	42439

Generated graph counts match OEIS A006820. The maximum at $n = 8$ is $K_{4,4}$. At $n = 10$ the maximum over all connected graphs of maximum degree ≤ 4 (89,402 graphs) equals the maximum over 4-regular graphs, consistent with edge monotonicity.

S3.3 Retracted preliminary analysis

A preliminary analysis of 43 curated terpenes preceded the COCONUT study. Its conclusions on mean difference and isoprene slope are retracted; its variance-ratio estimate was accurate.

quantity	curated, $n = 43$	COCONUT, $n = 6,121$	status
variance ratio	2.120, $p = 0.038$	2.045, $p = 1.4 \times 10^{-146}$	confirmed, within 4%
mean difference	-0.141 , $p = 0.106$	-0.599 , $d = -0.86$	retracted
isoprene slope	-0.118 /unit, $r = -0.351$	-0.046 /unit, $r = -0.110$	retracted

Terpenoid class means on COCONUT: monoterpenoid -0.030 ($n = 509$), sesquiterpenoid $+0.031$ ($n = 1,680$), diterpenoid -0.055 ($n = 1,898$), sesterterpenoid -0.108 ($n = 200$), triterpenoid -0.146 ($n = 1,834$).

S3.4 Normalisation exponents

Partial correlation with the boiling-point residual for $\log_2(Z)/n^k$, as the regression basis is enriched:

basis	$k = 1/3$	$k = 2/3$	$k = 1$
size terms only, linear	+0.444	+0.426	+0.360
size + $\log_2 Z$	+0.247	+0.220	+0.060
size + $\log_2 Z$ + nonlinear n	-0.087	-0.092	-0.096

S4. Software and reproducibility

Code, data and figures: github.com/guillaume-osmo/kumomlx

component	path
exact characteristic polynomial, dI_{pc} , $Z_{\max}$	<code>src/kumomlx/ipc.py</code>
derivation and measurements	<code>DESCRIPTORS.md</code>
upstream patch and Eigen C++ implementation	<code>rdkit-fix/</code>
$Z_{\max}$ enumeration	<code>rdkit-fix/zmax/</code>
COCONUT analysis	<code>paper/coconut.py</code> , <code>paper/coconut_stats.py</code>
figures	<code>paper/fig_*.py</code>

RDKit 2025.09.4; nauty 2.9.3, invoked as `geng -c -d4 -D4 n`.